COMP 511

Computer Programming Foundations II

Java – review of selected topics

The First program

Comments

Three forms of comments in Java

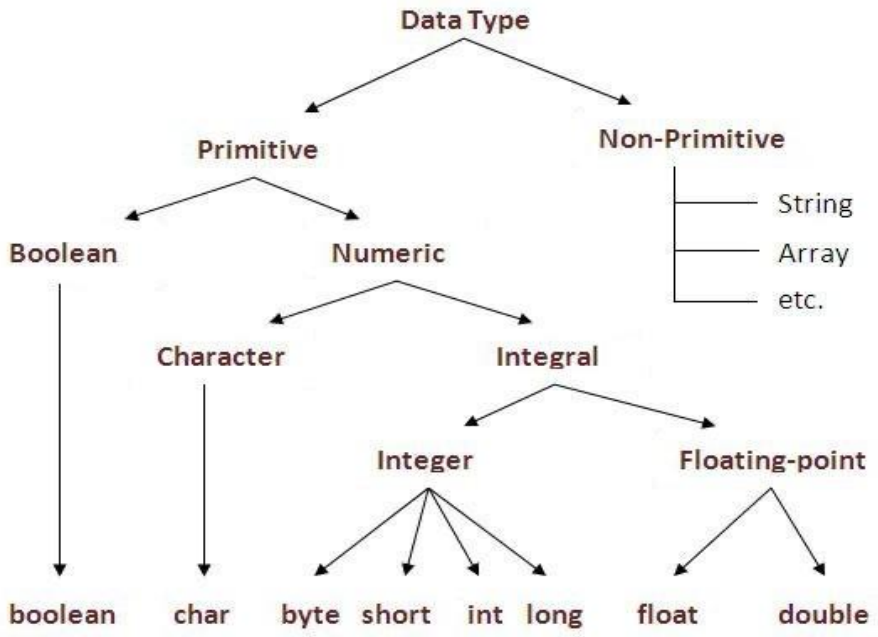
- /* … */
- // : Single line
- /** … **/ : Used to provide information to the Javadoc utility

Main

```
public static void main( String [ ] args )
```

Primitive Types

Primitive Type	What It Stores	Range
`byte`	8-bit integer	–128 to 127
`short`	16-bit integer	–32,768 to 32,767
`int`	32-bit integer	–2,147,483,648 to 2,147,483,647
`long`	64-bit integer	-2^{63} to $2^{63} - 1$
`float`	32-bit floating-point	6 significant digits (10^{-46}, 10^{38})
`double`	64-bit floating-point	15 significant digits (10^{-324}, 10^{308})
`char`	Unicode character	
`boolean`	Boolean variable	`false` and `true`



Declaration by providing variable name, its type, and optionally, its initial value.

Basic Operators

Assignment operators (=)

Binary arithmetic operators (* , + , …)

Unary operators (-- , ++)

Conditional Statements

Logical Operators

x	y	x && y	x || y	!x
false	false	false	false	true
false	true	false	true	true
true	false	false	true	false
true	true	true	true	false

IF Statement

```
if( expression )
    statement1
else
    statement2
next statement
```

Conditional Statements

While Statement

```
while( expression )
    statement
next statement
```

For Statements

```
for( initialization; test; update )
    statement
next statement
```

do Statement

```
do
    statement
while( expression );
next statement
```

break and continue

Conditional Statements

Switch Statement

```
switch( someCharacter )
{
  case '(':
  case '[':
  case '{':
    // Code to process opening symbols
    break;

  case ')':
  case ']':
  case '}':
    // Code to process closing symbols
    break;

  case '\n':
    // Code to handle newline character
    break;

  default:
    // Code to handle other cases
    break;
}
```

Methods

What is a method ?

- A *method* is similar to a function in other languages. The *method header* consists of the name, return type, and parameter list. The *method declaration* includes the body.
- Primitive types are passed using call-by-value
- The Return statement is used to return a value to the caller
- Void method ?

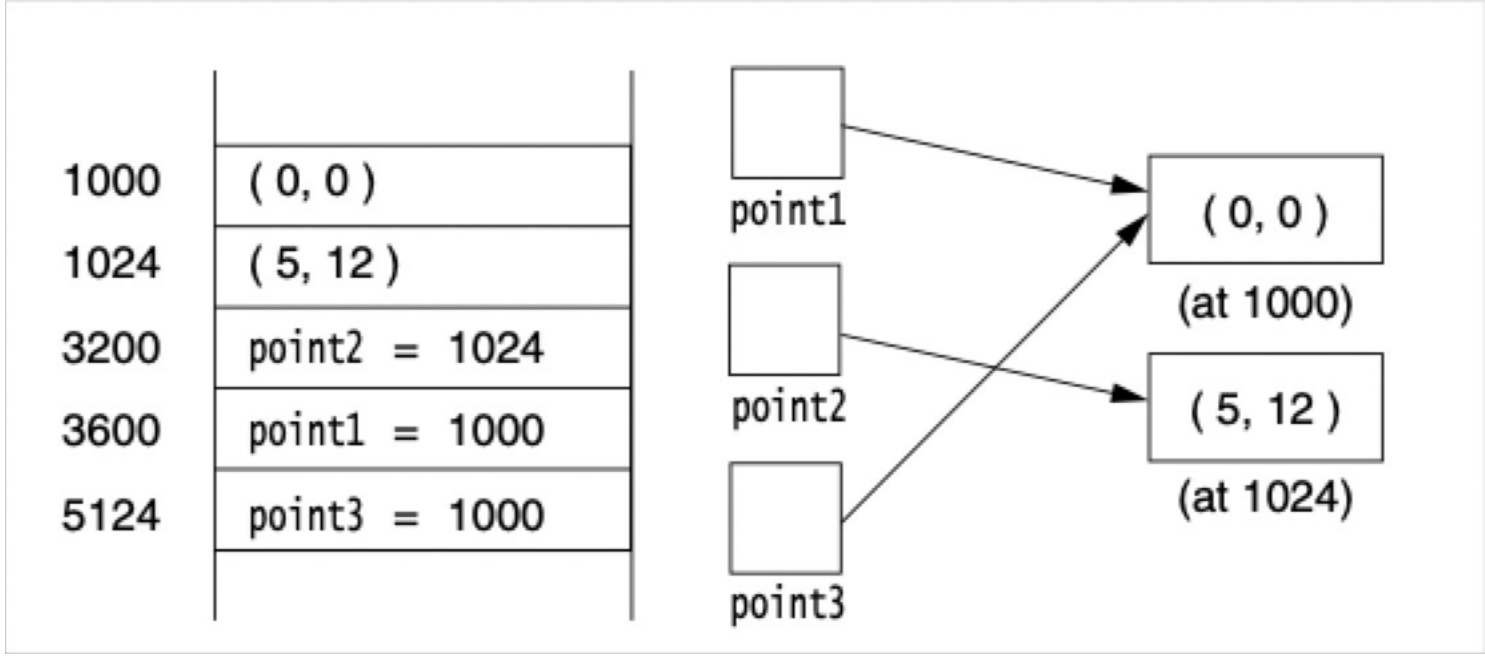
Overloading the methods names

```
int max( int a, int b, int c )
```

```
int max( int a, int b )
```

What is a Reference?

- A *reference variable* (often abbreviated as simply *reference*) in Java is a variable that somehow stores the memory address where an object resides.



Basics of Objects and References

The dot operator (.)

```
double theArea = theCircle.area( );
```

Declaration of objects

```
Button b = new Button( );
```

Garbage collection

In Java, unreferenced memory is automatically reclaimed.

Exceptions Handling

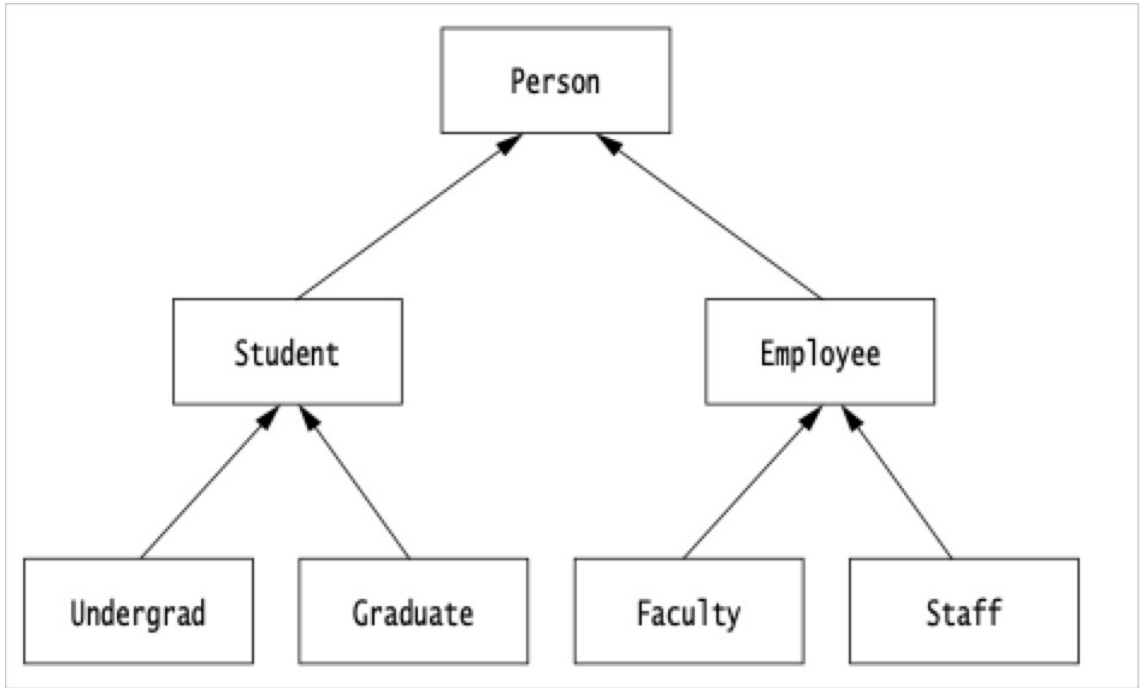
- *Exceptions* are used to handle exceptional occurrences such as errors.
- Use Try and Catch blocks

Common Exceptions

Standard Runtime Exception	Meaning
`ArithmeticException`	Overflow or integer division by zero.
`NumberFormatException`	Illegal conversion of `String` to numeric type.
`IndexOutOfBoundsException`	Illegal index into an array or `String`.
`NegativeArraySizeException`	Attempt to create a negative-length array.
`NullPointerException`	Illegal attempt to use a null reference.
`SecurityException`	Run-time security violation.
`NoSuchElementException`	Failed attempt to obtain “next” item

Inheritance

- *Inheritance* is the fundamental object-oriented principle that is used to reuse code among related classes
- Use the word extend
- super is used to call the base class constructor.
- Overriding methods

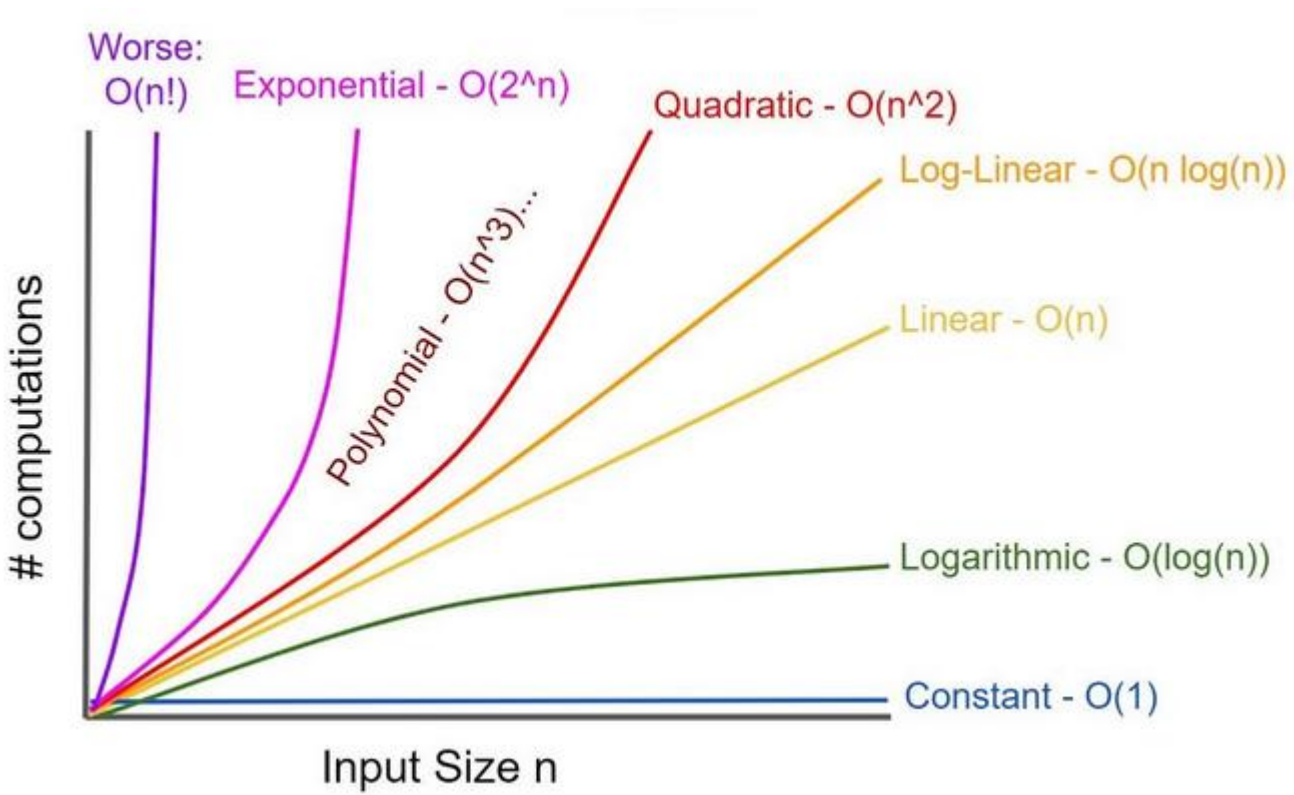


Algorithm Analysis?

- The amount of time that any algorithm takes to run almost always depends on the amount of input that it must process.

Big O Notation Summary

Notation	Type	Examples	Description
O(1)	Constant	Hash table access	Remains constant regardless of the size of the data set
O(log n)	Logarithmic	Binary search of a sorted table	Increases by a constant. If n doubles, the time to perform increases by a constant, smaller than n amount
O(< n)	Sublinear	Search using parallel processing	Performs at less than linear and more than logarithmic levels
O(n)	Linear	Finding an item in an unsorted list	Increases in proportion to n. If n doubles, the time to perform doubles
O(n log(n))	n log(n)	Quicksort, Merge Sort	Increases at a multiple of a constant
$O(n^2)$	Quadratic	Bubble sort	Increases in proportion to the product of n*n
$O(c^n)$	Exponential	Travelling salesman problem solved using dynamic programming	Increases based on the exponent n of a constant c
O(n!)	Factorial	Travelling salesman problem solved using brute force	Increases in proportion to the product of all numbers included (e.g., 1*2*3*4…)



Let’s program a basic program…

We want a Java program to check the grade of the student at a class. The grade scale should be pre-defined e.g. the grade of someone who gets 90 is A.